



15 Linux Problems Every DevOps Engineer Should Know How to Fix

Linux is the backbone of modern DevOps. Whether you're managing Kubernetes nodes, EC2 instances, or on-premise servers, troubleshooting Linux efficiently is a must-have skill.

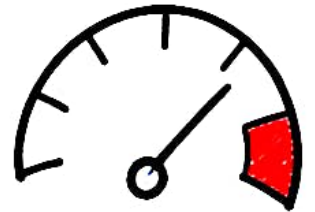
Here are 15 common Linux production issues along with the commands that help diagnose and resolve them.



① Server Is Slow

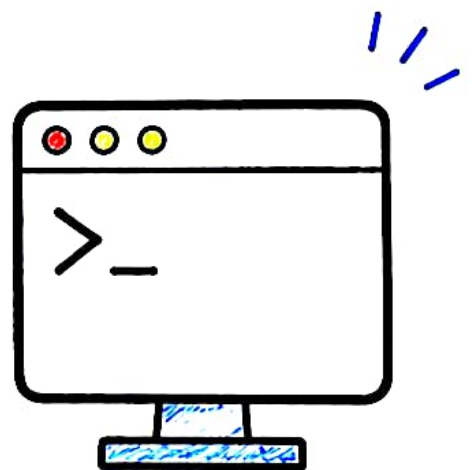
Symptoms

- High load average
- Slow response
- Applications lagging



Commands

- `top`
- `htop`
- `vmstat 1`
- `iostat -x 1`
- `sar -u 1 5`



② Disk Space Is Full

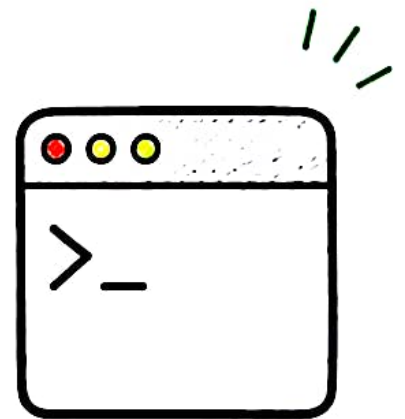


Symptoms

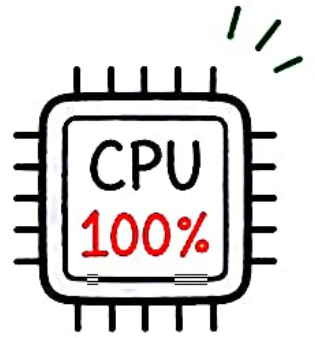
- “No space left on device”
 - Logs or applications stop working
-

Commands

- `df -h`
- `du -sh /* | sort -hr`
- `ncdu`
- `lsblk`
- `journalctl --vacuum-time=7d`



③ High CPU Usage

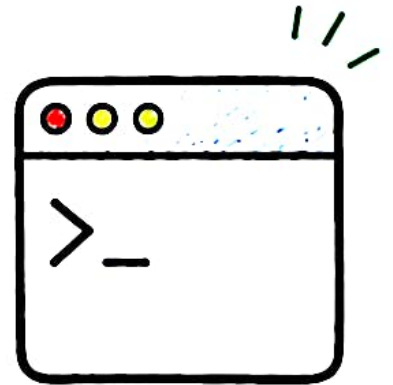


Symptoms

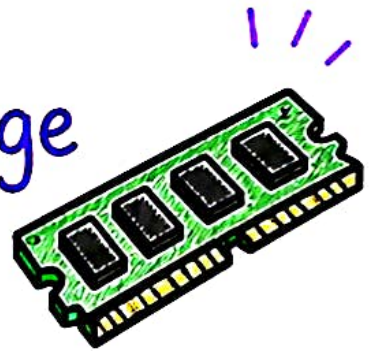
- CPU reaches 100%
 - System becomes unresponsive
-

Commands

- `top`
- `ps aux --sort==%cpu | head`
- `mpstat -P ALL`
- `sar -u 1 5`
- `pidstat -u 1`



④ High Memory Usage

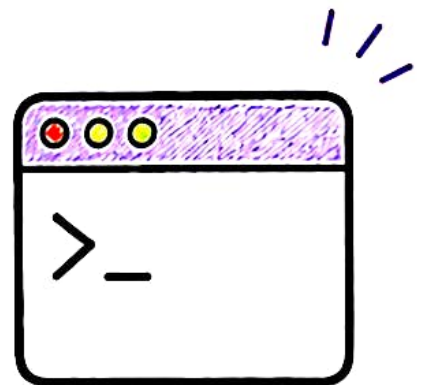


Symptoms

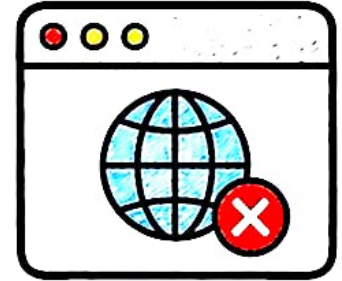
- Out Of Memory (OOM)
 - Slow system performance
-

Commands

- `free -h`
- `vmstat`
- `cat /proc/meminfo`
- `ps aux ==sort=-%mem | head`
- `pmap -x <PID>`



⑤ Website or Application Not Reachable



Symptoms

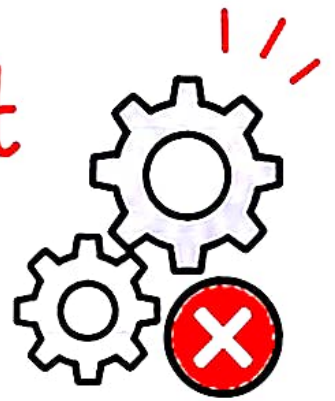
- Users cannot access the application

Commands

- `ping <host>`
- `curl -I http://<host>`
- `traceroute <host>`
- `ss -tulpn`
- `systemctl status <service>`



⑥ Service Won't Start

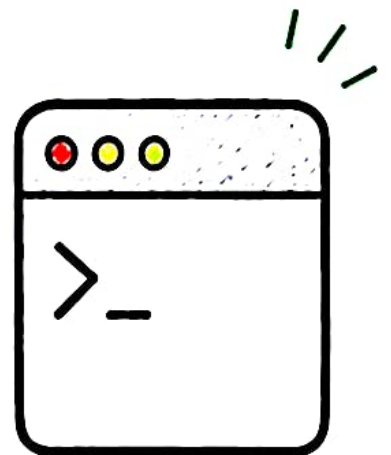


Symptoms

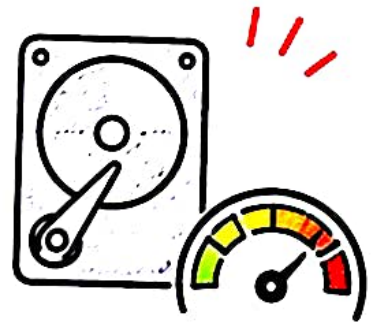
- Service fails or keeps crashing

Commands

- `systemctl status <service>`
- `journalctl -u <service> -xe`
- `systemctl cat <service>`
- `ps aux | grep <service>`
- `dmesg | tail`



⑦ Disk I/O Is High



Symptoms

- Slow disk operations
- High wait time

Commands

- `iostat -x 1`
- `dstat -d`
- `pidstat -d 1`
- `iostat`
- `blktrace /dev/sdX`



⑧ Port Is Not Listening

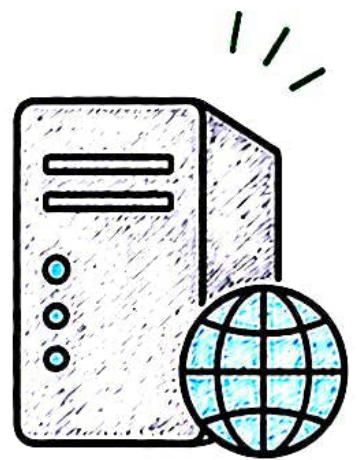


Symptoms

- Application isn't accepting connections

Commands

- `ss -tulpn | grep <port>`
- `netstat -tulpn | grep <port>`
- `lsof -i :<port>`
- `firewall-cmd --list-ports`
- `ufw status`



⑨ DNS Resolution Fails

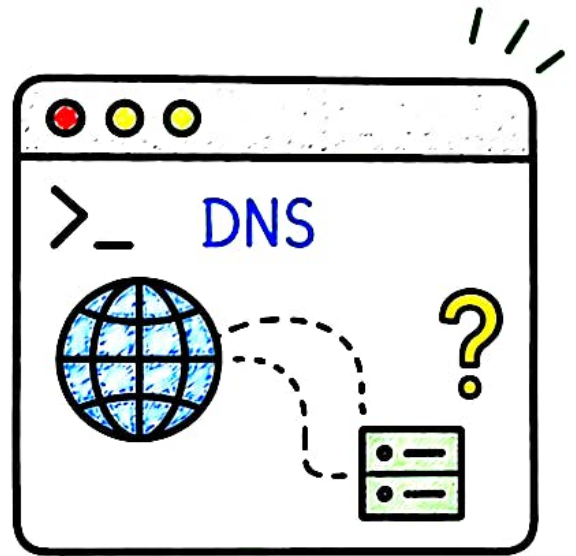


Symptoms

- Unable to resolve hostnames

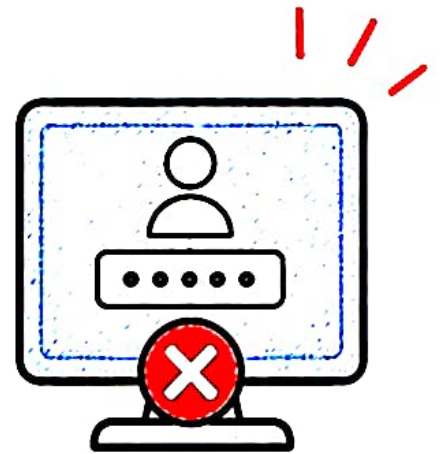
Commands

- `dig <domain>`
- `nslookup <domain>`
- `cat /etc/resolv.conf`
- `systemd-resolve --status`
- `ping 8.8.8.8`



Tip: If ping 8.8.8.8 works but domain name doesn't resolve, the issue is DNS server or configuration.

⑩ Login Failure

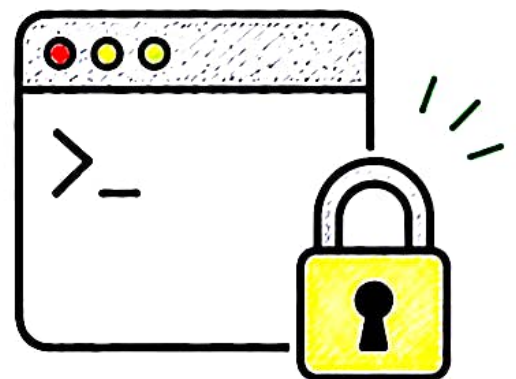


Symptoms

- Unable to log in via SSH

Commands

- `journalctl -xe`
- `tail -f /var/log/auth.log`
- `last -xe`
- `faillock --user <user>`
- `grep -i fail /var/log/*`

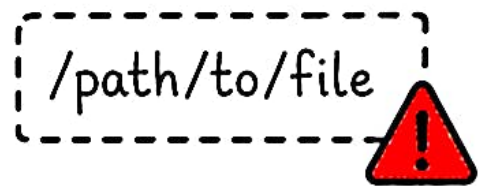


11 File or Directory Not Found



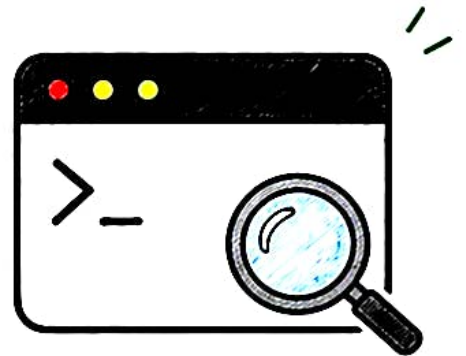
Symptoms

- Missing files
- Path errors



Commands

- `find / -name <file> 2>/dev/null`



- `locate <file>`



- `ls -lah <path>`



- `df -H`



- `stat <file>`



12 Permission Denied



Symptoms

- Cannot access files or directories

Commands

- `ls -l <file>`

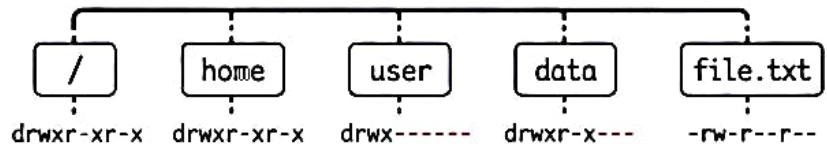
```
-rw-r--r-- 1 user group 1024 May 20 file.txt
```

- `id`



```
uid=1000(user)
gid=1000(user)
groups=1000(user),10(wheel)
```

- `namei -l <path>`



- `getfacl <file>`

```
# file: file.txt
# owner: user
# group: user
user::rw-
group::r--
other::r--
user:john:r--
group:dev:r-x
mask::r-x
other::r--
```



- `sudo -l`

```
Matching Defaults entries for user on this host:
env_reset, mail_badpass,
secure_path=/usr/local/sbin:\
/usr/local/bin:/usr/sbin:/usr/bin

User user may run the following commands on this host:
(ALL) NOPASSWD: /usr/bin/systemctl restart apache2
```



13 Log Files Are Huge



Symptoms

- Disk fills up because of logs

Commands

- `du -sh /var/log/*`

```
512M    /var/log/secure
256M    /var/log/messages
1.2G    /var/log/app.log
2.0G    /var/log/
```



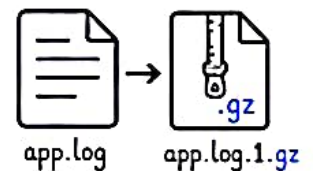
- `journalctl --disk-usage`

```
Archived and active journals take up
2.5G on disk.
```



- `logrotate /etc/logrotate.conf`

```
rotating pattern: /var/log/*.log
creating new /var/log/app.log
old log saved as /var/log/app.log.1.gz
```



- `tail -f <logfile>`

```
2025-05-20 10:15:21 INFO    Starting service...
2025-05-20 10:15:22 INFO    Service started
2025-05-20 10:15:25 WARN    High memory usage
2025-05-20 10:15:26 ERROR   Failed to connect DB
2025-05-20 10:15:27 INFO    Retrying...
```



- `grep -i error <logfile>`

```
2025-05-20 10:15:26 ERROR   Failed to connect DB
2025-05-20 10:16:03 ERROR   Timeout occurred
2025-05-20 10:17:41 ERROR   Invalid user token
```



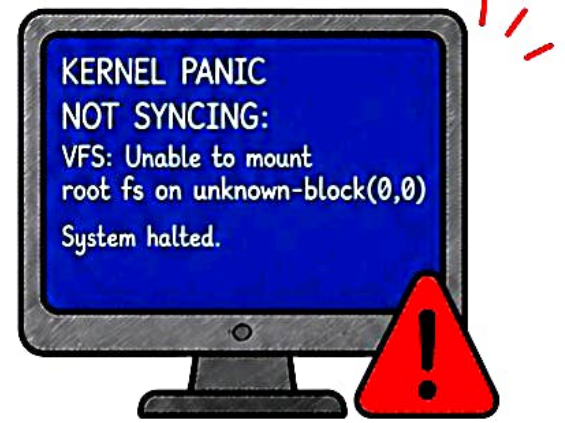
Tip: Configure logrotate properly and monitor log size regularly to prevent disk space issues.



14 Kernel Panic or System Crash

Symptoms

- Unexpected reboot
- System crash



Commands

- `dmesg -T | tail -50`

```
[ 128.123456] CPU: 3 PID: 0 Comm: swapper/3 Not tainted 5.15.0
[ 128.123789] Hardware name: Dell Inc. PowerEdge R740xd, BIOS 2.9.0
[ 128.124000] RIP: 0010:panic+0x101/0x2a0
[ 128.124210] Kernel panic - not syncing: VFS: Unable to mount
root fs on unknown-block(0,0)
```



- `journalctl -k -b -1`

```
-- Logs begin at Fri 2025-05-19 08:23:11 IST, end at Sat 2025-05-20 07:10:02 IST. --
May 19 08:24:11 server kernel: Kernel panic - not syncing: Attempted to kill
init! exitcode=0x0000000b
May 19 08:24:11 server kernel: CPU: 1 PID: 1 Comm: init Not tainted 5.15.0
May 19 08:24:11 server kernel: Call Trace:
...
```



- `cat /proc/uptime`

```
1234567.89      987654.32
  System uptime   Idle time
   (seconds)      (seconds)
```



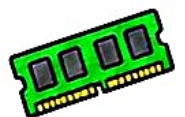
- `lsmod`

Module	Size	Used by
ext4	147456	1
crc32c	16384	2
xfs	188416	1
libcrc32c	16384	3
ata_generic	20480	0



- `free -hq`

	total	used	free	shared	buff/cache	available
Mem:	15G	4.2G	6.1G	512M	4.7G	9.8G
Swap:	4.0G	512M	3.5G	-	-	3.2G



Tip: Check recent kernel logs and hardware status. Repeated crashes may indicate hardware failure, driver issues, or memory problems.

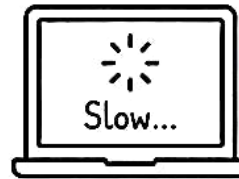


15 Network Is Slow



Symptoms

- High latency
- Slow application response



Commands

- ping <host>

```
$ ping google.com
PING google.com (142.250.72.206) 56(84) bytes of data:
64 bytes from 142.250.72.206: icmp_seq=1 ttl=115 time=120 ms
64 bytes from 142.250.72.206: icmp_seq=2 ttl=115 time=118 ms
64 bytes from 142.250.72.206: icmp_seq=3 ttl=115 time=119 ms
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 118.321/119.321/120.654/0.986 ms
```



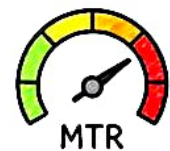
- traceroute <host>

```
$ traceroute google.com
traceroute to google.com (142.250.72.206), 30 hops max
 1  192.168.1.1      1.123 ms  1.045 ms  1.012 ms
 2  10.0.0.1         5.234 ms  5.210 ms  5.198 ms
 3  100.64.1.1       15.234 ms 15.120 ms 15.098 ms
 4  142.250.72.206   119.876 ms 119.654 ms 119.432 ms
```



- mtr <host>

```
$ mtr google.com
HOST: myserver
 1. 192.168.1.1      0.0% 10 1.1 1.2 0.9 1.6 0.2
 2. 10.0.0.1        0.0% 10 5.1 5.3 4.7 6.1 0.4
 3. 100.64.1.1      0.0% 10 15.2 15.3 14.8 16.2 0.4
 4. 142.250.72.206  0.0% 10 119.6 119.7 118.9 120.6 0.5
```



- ip -s link

```
$ ip -s link
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP
RX:  bytes  packets  errors  dropped  overrun  mcast
    1234567890 1234567  0      12      0      12345
TX:  bytes  packets  errors  dropped  carrier  collsns
    987654321 987654  0       6      0       0
```



- ethtool <interface>

```
$ ethtool eth0
Settings for eth0:
Supported ports: [ TP ]
Supported link modes: 1000baseT/Full
Speed: 1000Mb/s
Duplex: Full
Auto-negotiation: on
Link detected: yes
```



Tip: Check bandwidth usage, packet loss, latency, and interface errors to identify network bottlenecks.





Golden Rules for Linux Troubleshooting

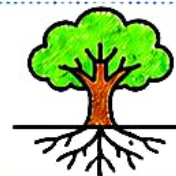
1

Observe the symptoms before acting.



2

Identify the root cause instead of guessing.



3

Fix the issue methodically.



4

Verify the resolution.



5

Take preventive measures to avoid recurrence.



Pro Tips



Always check logs first.



Don't **kill processes** without understanding why they're consuming resources.



Document the root cause and the fix.



Learn from every production incident.

